Starbucks Customer Survey Analysis

This report converts the customer survey into business-facing insight on customer composition, brand experience, loyalty patterns, promotion effectiveness, and actionable targeting opportunities. It also documents the key methodological decisions used in processing and interpreting the survey so that recommendations can be used appropriately and validated further where needed.

Dataset: 122 survey responses

Scope: demographics, visit behavior, spend, membership, satisfaction, promotions, continuation intent

Use case: strategy guidance, campaign design, experiment planning, and future validation

69.7%
Aged 20–29

91.7%
Continuation intent among members

3.80 / 5
Average promotion importance

0
Exact duplicate rows

Executive summary

1. The customer base in this survey is heavily concentrated in younger adults, particularly ages 20–29, with a slight female skew and a large lower-income cohort.
2. Starbucks appears to be positioned primarily as an **experience-led premium brand**: ambiance, service, and quality score more strongly than price.

3. Promotions still matter meaningfully in purchase decisions despite the premium positioning, especially for younger and lighter users.
4. Customers with membership are associated with higher frequency, higher spend, stronger experience scores, and stronger continuation intent.
5. The strongest observed correlates of continuation intent are perceived price/value acceptability, spend, visit frequency, quality, hangout suitability, and membership status.
6. Social media is the broadest promotion channel overall, while owned channels such as the Starbucks app/site appear more relevant for higher-value and member-heavy audiences.
7. A segment-based strategy is still useful for campaign design, but the segment structure should be treated as **directional** rather than definitive and should be validated against transaction and CRM data.

Data quality and limitations

Quality checks: the dataset contains 122 rows, no raw nulls, and no exact duplicate rows.

Retained edge cases: responses such as `Never buy`, `Unknown / invalid`, and `Zero` spend were retained and standardized rather than dropped. This preserves transparency and avoids arbitrary exclusions, but it also means results should be interpreted with business context in mind.

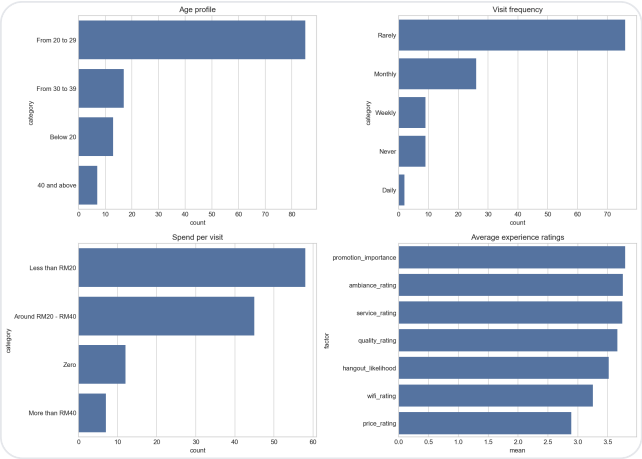
Main limitations

- This is a survey sample, not a transaction-level population dataset.
- Continuation intent is self-reported rather than observed repeat purchase behavior.
- Age and income are captured in broad bands, which limits precision.
- Some conclusions, especially around segmentation, are more suitable for hypothesis generation than for definitive classification.

Customer profile overview

- 53.3% female, 46.7% male
- 69.7% aged 20–29
- 50.0% employed
- 34.4% student
- 58.2% below RM25,000 income
- Most common visit pattern: rare visits
- Most common spend: less than RM20 per visit
- Main usage modes: take-away and dine-in

The customer picture is that of a predominantly young, relatively budget-constrained audience who still engages with Starbucks for both convenience and café/social use occasions. This matters because it helps explain why premium experience scores can coexist with meaningful promotion sensitivity.



Age mix, visit frequency, spend profile, and average experience scores.

Satisfaction and brand experience

The strongest experience dimensions in the survey are ambiance, service, and product quality. Price is consistently the weakest-rated dimension. The combination suggests that Starbucks is being evaluated primarily through a premium-experience lens rather than through low-price competitiveness.

That distinction is commercially important. The survey does not suggest that customers simply want cheaper coffee. Instead, it suggests that loyalty is stronger when customers feel the experience is worth the price. In other words, value-for-money matters more than price alone.

Membership and customer value

Status	Respondents	Avg frequency	Avg spend	Retention %	Avg experience	Promotion importance
Non-member	62	1.06	1.15	62.9	3.38	3.68
Member	60	1.62	1.63	91.7	3.80	3.92

The pattern is clear: members in this sample tend to be more valuable. They visit more often, spend more, report stronger overall experience, and express much higher continuation intent.

Interpret carefully: this survey shows association, not causation. It does not prove that membership itself created the higher value. It may also be true that already-loyal customers are more likely to sign up.

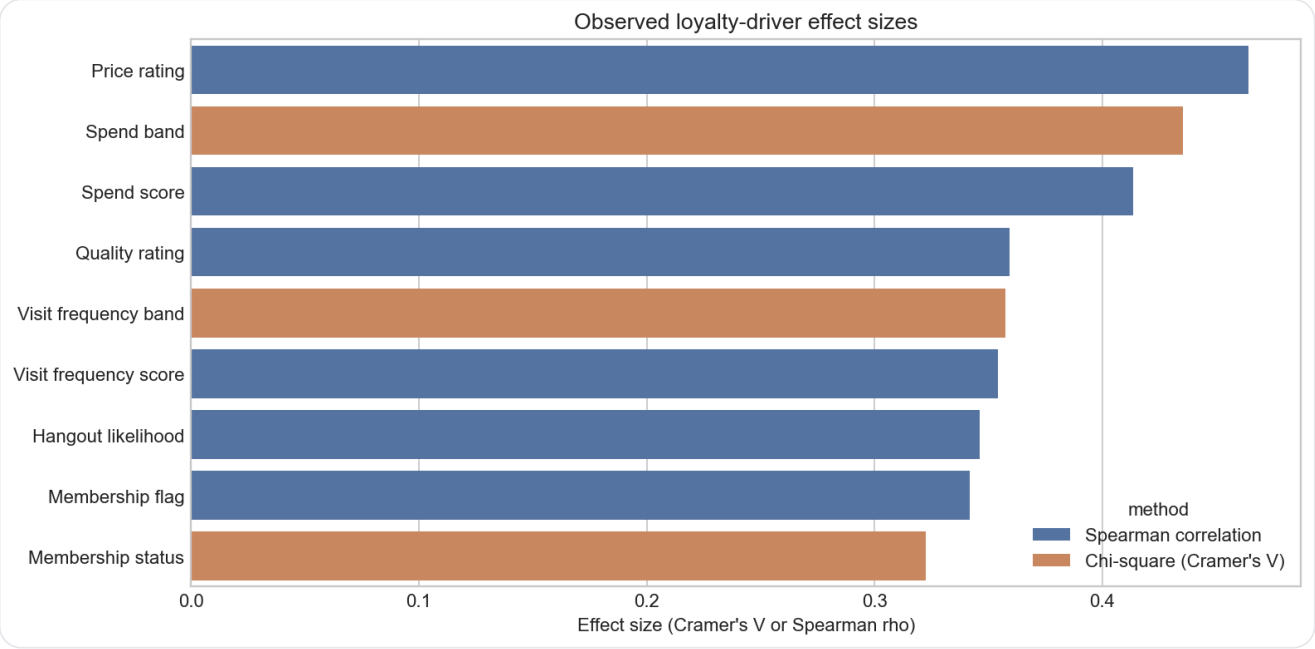
Recommended implication: treat membership as a promising *testable loyalty lever*. A sensible next step is to run app-first membership conversion experiments for high-potential non-members and measure whether conversion changes repeat visits and spend.

Loyalty driver analysis

The survey analysis tested several potential relationships with continuation intent using chi-square tests for categorical relationships and Spearman correlations for ordered or numeric relationships. The table below shows the compact evidence summary.

Relationship tested	Method	Effect size	p-value	Interpretation	Significant at 0.05
Spend band vs continuation intent	Chi-square (Cramer's V)	0.435	0.0000	moderate association	Yes
Membership status vs continuation intent	Chi-square (Cramer's V)	0.322	0.0004	moderate association	Yes
Visit frequency band vs continuation intent	Chi-square (Cramer's V)	0.358	0.0036	moderate association	Yes

Relationship tested	Method	Effect size	p-value	Interpretation	Significant at 0.05
Price rating vs continuation intent	Spearman correlation	0.464	0.0000	moderate association	Yes
Spend score vs continuation intent	Spearman correlation	0.414	0.0000	moderate association	Yes
Quality rating vs continuation intent	Spearman correlation	0.359	0.0000	moderate association	Yes
Visit frequency score vs continuation intent	Spearman correlation	0.354	0.0001	moderate association	Yes
Hangout likelihood vs continuation intent	Spearman correlation	0.346	0.0001	moderate association	Yes
Membership flag vs continuation intent	Spearman correlation	0.342	0.0001	moderate association	Yes



Observed effect sizes for the strongest relationships with continuation intent.

Interpretation

- **Spend band, membership status, and visit frequency band** all show statistically meaningful associations with continuation intent.
- **Price rating / value acceptability** is the strongest observed attitudinal correlate in the sample.

- **Quality, hangout suitability, and frequency** also matter meaningfully, reinforcing the idea that Starbucks wins when it functions as both a premium beverage brand and a useful social/work setting.

Promotion effectiveness

Promotions remain commercially relevant in this customer base. Social media is the broadest channel overall, making it the best general-purpose top-of-funnel or reach vehicle in this dataset. Owned channels such as the Starbucks website or app are more aligned with higher-value and member-heavy audiences, while word of mouth remains relevant for lighter and younger groups.

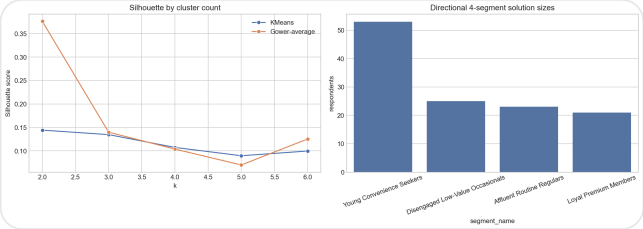
This supports a split approach:

- **Social-first outreach** for younger, convenience-led, and occasional customers
- **Owned CRM/app communication** for members and higher-frequency customers
- **Referral and peer influence mechanics** where word of mouth appears strong

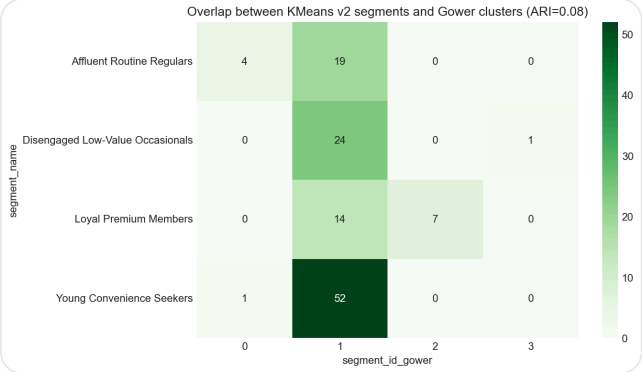
Segmentation methodology and interpretation

The survey was segmented using a four-group clustering framework to create business-usable customer profiles. This is useful for marketing design, but there is an important methodological caveat: when the clustering was checked using a mixed-data sensitivity approach, the detailed four-group structure was not strongly reproduced. Cross-method agreement was weak (ARI ≈ 0.08).

What this means: the segment framework is still helpful for campaign design and internal planning, but it should be treated as a directional strategy tool rather than a definitive natural market segmentation.



Cluster validation summary.



Cross-method overlap between the main segmentation and the mixed-data sensitivity check.

Directional segment profiles

Segment	Size	% of sample	Member rate %	Retention %	Avg frequency	Avg spend	Top visit mode	Top visit frequency
Young Convenience Seekers	53	43.4	37.7	77.4	1.13	1.15	Take away	Rarely
Disengaged Low-Value Occasionals	25	20.5	8.0	48.0	0.80	1.04	Dine in	Rarely
Affluent Routine Regulars	23	18.9	82.6	87.0	1.96	1.91	Dine in	Monthly
Loyal Premium Members	21	17.2	90.5	100.0	1.81	1.81	Dine in	Rarely

How to use these segments

These segments are best treated as targeting lenses or campaign hypotheses. They are useful because they translate survey patterns into actionable marketing and experience strategies, even if the exact boundaries between groups are not perfectly sharp.

- **Young Convenience Seekers:** use for quick-treat, convenience-led, social-first offers and app sign-up nudges.
- **Disengaged Low-Value Occasionals:** use for trial reactivation, entry-price bundles, and clearer value communication.
- **Affluent Routine Regulars:** use for habit reinforcement, CRM targeting, and convenience/service improvements.
- **Loyal Premium Members:** use for exclusives, tier benefits, referral programs, and advocacy mechanics.

Data-processing appendix

The appendix below documents the major data preparation choices that can materially affect survey interpretation, including how inconsistent labels and edge-case responses were handled.

Processing step	What changed	Details	Impact on analysis
Row count	Checked exact duplicate rows on the raw file	122 rows total; 0 duplicates found	No rows removed
Column renaming	Renamed survey questions to business-readable snake_case fields	All 21 columns renamed in analysis layer	Improves readability only
Visit mode standardization	Standardized inconsistent visit-mode labels	Mapped never/Never/Never buy -> Never buy; NA and I dont like coffee -> Unknown / invalid	7 rows retained and flagged as non-active / invalid-like
Spend standardization	Standardized spend-band labels	Mapped zero -> Zero and kept Zero as a valid explicit response	12 rows retained with spend score 0
Multi-select parsing	Split semicolon-separated multi-select responses	Parsed purchase_items into 11 binary flags; promotion_channels into 10 binary flags	Original text retained alongside parsed lists
Ordinal encoding	Encoded ordered categories into interpretable scores	Age, income, visit frequency, spend, duration, and distance converted to ordinal scores	Used for comparisons and clustering; original labels retained for reporting

Processing step	What changed	Details	Impact on analysis
Ratings handling	Converted 1-5 ratings to numeric values	quality, price, promotion importance, ambiance, wifi, service, hangout	No imputation needed because raw fields were complete
Missing values / NA	Checked nulls and explicit NA-like strings	No raw nulls; NA-like response in visit mode retained as Unknown / invalid	No rows dropped for missingness
Edge-case response treatment	Retained but documented unusual responses	Never buy / Unknown-like / Zero spend rows kept to avoid arbitrary exclusions	Interpret results directionally and review sensitivity against business context

Recommendations

1. **Protect the premium experience** by maintaining strong ambiance and service, since those remain central to brand strength.
2. **Improve value communication** through bundles, occasion-based offers, and premium-worth framing rather than relying only on discounting.
3. **Keep social media as the main broad-reach channel**, especially for younger and lighter users.
4. **Use membership conversion as an experiment**, not as a presumed causal engine of loyalty.
5. **Use segment-specific messaging and offers**, but treat the segment model as directional until validated with behavioral data.
6. **Validate key findings with transaction and CRM data** before making large strategic commitments.

Measurement plan and next steps

KPIs to track

- Visit frequency
- Average spend per visit
- Membership conversion rate
- Repeat purchase / continuation rate

- Promotion redemption rate
- Social-to-store or social-to-app conversion
- Satisfaction by factor: quality, price/value, service, ambiance
- Movement between customer targeting groups over time

Recommended next steps

1. Run membership-conversion experiments among high-potential non-members.
2. Validate the survey-led findings against loyalty, transaction, and CRM data.
3. Repeat the segmentation on a larger and fresher sample.
4. Improve the next survey by separating price level from value-for-money and by measuring actual repeat behavior more directly.